APPLIED MATHEMATICS I: YEAR ONE

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1. Logic
2. Algebraic structure
3. Number Theory
4. Polynomials
5. Equations and Inequalities
6. Linear Transformation, Matrices and determinant of matrices
7. Geometry and Trigonometry
8. Calculus
9. Probability and Statistics

Introduction

Introduction

Logic is the study of the principles of reasoning. It is concerned with what is true and how we can know whether something is true.

Consider the following compound sentence:

All human beings are mortal

Sakabaka is not a human being.

Therefore, Sakabaka is not mortal.

Introduction

Here the conclusion is not true since it is said that all human being are mortal and is known that Sakabaka is not a human being. So, Sakabaka can, for example, be a dog which is mortal!

Mathematical logic helps us to understand how this compound sentence is true or false. Use of the logical style helps us to recognize patterns easily, as well as connections between seemingly meaningless content.

Objectives

After completing this unit, the learners should be able to:

- *Distinguish between statement and proposition*
- *Convert into logical formula composite propositions and vice versa*
- *Construct the truth table of a composite proposition*
- *Recognize the most often used tautologies*
- *Use quantifiers*
- *Determine whether a given argument is valid or fallacy*

Definition and examples

A proposition

A proposition (statement or verbal assertion) is a sentence which is either true or false

Notation

1. The two truth values of proposition are **true** and **false** and are denoted by the symbols **T** and **F** respectively.
2. Occasionally they are also denoted by the symbols **1** and **0** respectively.
3. Statements will be denoted by small letters such as **p, q, r, ...**

Examples

From the following expression, give your answer by true or false where possible and hence give the expressions which are propositions

1. Kampala is in Rwanda.
2. $\sqrt{16} = 4$.
3. When were you born?
4. Rwanda is African country.
5. $x + y = 9$.
6. Write down the names of Rwandan president.
7. All the above sentences are right.

Solution

1. Kampala is in Rwanda. **It is a false proposition.**
2. $\sqrt{16} = 4$. **It is a true proposition.**
3. When were you born? **It is not a proposition.**
4. Rwanda is African country.**It is a true proposition.**
5. $x + y = 9$. **It is not a proposition.**
6. Write down the names of Rwandan president. **It is not a proposition.**
7. All the above sentences are right.**It is a false proposition.**

Exercise

Are these propositions? If any, give their true values

1. Uganda is a member of East African Community.
2. The sun is shining.
3. Paris is in England.
4. Come to class!
5. The sum of two prime numbers is even.
6. It is not true that Uganda is in Europe.

Truth table

Truth table of compound statements

1. We can always summarize the truth values of compound statement in a table called truth table.
2. If the compound statement contains **n** distinct components, we need to consider 2^n possible combinations of truth values in order to obtain the truth table.

For one proposition **p** we have 2 possible combinations thus, the truth table is

p
T
F

Or

p
1
0

For 2 propositions **p and q** we have 2^2 possible combinations thus, the truth table is

p	q
T	T
T	F
F	T
F	F

Or

p	q
1	1
1	0
0	1
0	0

For 3 propositions **p, q and r** we have 2^3 possible combinations thus, the truth table is

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Or

p	q	r
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0
0	0	1
0	0	0

Propositional Logic

Logical connectives

1. We can formulate compound statements by using connectors
2. We introduce three symbols that are used to build more complicated statement out of simpler ones.
3. We have the symbols $\neg$, $\wedge$ and $\vee$ to denote respectively **not, and, or.**

The negation of a statement

1. The negation of a statement by introducing the word "not" denoted by prefixing the statement has opposite truth value from the statement.
2. It is denoted by $\neg p$ or $\sim p$ or $\bar{p}$
3. From this definition, it follows that if p is true, then $\neg p$ is false and vice-versa.

Examples

1. The truth value of $\neg p$ relative to p is given in the following table.

p	$\neg p$
1	0
0	1

2. Complete the following truth table

p	q	r	$\neg p$	$\neg q$	$\neg r$
1		1		0	
	1	0	0		
1	0	1			
1	0				1
	1	1	1		
0	1	0			
0				1	0
0	0	0			

The conjunction (AND)

1. The conjunction of two statements**p** and **q** is denoted $p \wedge q$.
2. It has the truth value **true** whenever both **p** and **q** have the truth value **true**; otherwise it has the truth value **false**.

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

Examples

Let p and q be propositions. Construct the truth table for

a) $p \wedge q$

b) $\neg p \wedge q$

c) $(\neg p \wedge q) \wedge \neg p$

Solution

p	q	$p \wedge q$	$\neg p$	$\neg p \wedge q$	$(\neg p \wedge q) \wedge \neg p$
1	1	1	0	0	0
1	0	0	0	0	0
0	1	0	1	1	1
0	0	0	1	0	0

Exercises

Let p and q be two propositions. Construct the truth table of

1. $p \wedge q$
2. $\neg p \wedge q$
3. $p \wedge \neg q$
4. $\neg(p \wedge q)$
5. $\neg q \wedge (\neg p \wedge q)$

The disjunction (OR)

1. The The disjunction of two statements**p** and **q** is denoted $p \vee q$.
2. It has the truth value **false** only when **p** and **q** have truth value **false**, otherwise it has the truth value **true**.
3. If p and q are two propositions then the truth table of $p \vee q$ is given by the following table:

p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0

Examples

Determine the truth value of each of the following statements

1. Paris is in France or $4 + 4 = 8$
2. $4 + 4 = 9$ or $5 + 8 = 11$
3. The first president of United States of America is George Washington or was inaugurated in 1879.
4. Kigali is the capital of Burundi or $1 + 1 = 2$
5. The French revolution started in 1789 or ended in 1799
6. CO_2 is chemical formula of water or H_2O is chemical formula of carbon dioxide
7. m^2 is the unit of area or kg is one of the units of weight.

Examples

SOLUTION

1. Paris is in France or $4 + 4 = 8$. **True**
2. $4 + 4 = 9$ or $5 + 8 = 11$.**False**
3. Kigali is the capital of Burundi or $1 + 1 = 2$. **True**
4. The French revolution started in 1789 or ended in 1799.
5. CO_2 is chemical formula of water or H_2O is chemical formula of carbon dioxide.**False**
6. m^2 is the unit of area or kg is one of the units of weight. **True**

From propositions p, q and r, construct the truth table for

a) $(p \vee q) \wedge r$

b) $p \vee (\neg q)$

c) $(p \wedge q) \vee (\neg p)$

Solution

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$\neg q$	$p \vee (\neg q)$	$p \wedge q$	$\neg p$	$(p \wedge q) \vee (\neg p)$
1	1	1	1	1	0	1	1	0	1
1	1	0	1	0	0	1	1	0	1
1	0	1	1	1	1	1	0	0	0
1	0	0	1	0	1	1	0	0	0
0	1	1	1	1	0	0	0	1	1
0	1	0	1	0	0	0	0	1	1
0	0	1	0	0	1	1	0	1	1
0	0	0	0	0	1	1	0	1	1

Tautology and Contradiction

Tautology

1. A tautology is a statement formula that is always true regardless of the truth values of the individual statements substituted for its statement variables.
2. A statement whose formula is a tautology is called a tautological statement.

Contradiction

1. A contradiction is a statement formula that is always false regardless of the truth values of the individual statements substituted for its statement variables.
2. A statement whose formula is a contradiction is called contradictory statement.

Example

The statement formula $p \vee \neg p$ is a tautology and the statement $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee (\neg p)$	$p \wedge (\neg p)$
1	0	1	0
0	1	1	0

Exercises

For below compound statements, indicate which is tautology, contradiction or neither.

1. $p \wedge \neg(p \wedge q)$
2. $\neg q \wedge (q \wedge r)$
3. $(p \wedge q) \wedge \neg(p \vee q)$
4. $(p \vee r) \vee \neg r$

Logical equivalences

Two statements are said to be logically equivalence if they have the same truth values.

Test for logical equivalence

To test whether two statements formulas **p** and **q** are logically equivalent:

1. Construct the truth table for **p**,
2. Construct the truth table for **q** using the same statement variables for identical component statement,
3. Check each combination of truth values of the statement variables to see whether the truth values of **p** is the same as the truth values of **q**.
4. If in each row truth value of **p** is the same as the truth values of **q**, then **p** and **q** are logically equivalent.
5. If in same row **p** has a different truth value from **q**, then **p** and **q** are not logically equivalent.

Examples:Verify that the following propositions are logical equivalent

a) $p \wedge q$ *and* $\neg(\neg p \vee \neg q)$

b) $p \vee q$ *and* $\neg(\neg p \wedge \neg q)$

a)

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee \neg q$	$\neg(\neg p \vee \neg q)$
1	1	0	0	1	0	1
1	0	0	1	0	1	0
0	1	1	0	0	1	0
0	0	1	1	0	1	0

Since the truth tables of $p \wedge q$ and $\neg(\neg p \vee \neg q)$ are identical then the propositions are equivalent.

b)

p	q	$\neg p$	$\neg q$	$p \vee q$	$\neg p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
1	1	0	0	1	0	1
1	0	0	1	1	0	1
0	1	1	0	1	0	1
0	0	1	1	0	1	0

Since the truth tables of $p \vee q$ and $\neg(\neg p \wedge \neg q)$ are identical then the propositions are equivalent.

Logical equivalences Theorem

Given any statement variables **p, q** and **r**; a tautology **t** and a contradiction **c** the following logical equivalences hold:

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge t \equiv p$	$p \vee c \equiv p$
Negation laws	$p \wedge (\neg p) \equiv c$	$p \vee (\neg p) \equiv t$

Double negation laws	$\neg(\neg p) \equiv p$	-
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
De Morgan laws	$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$	$\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$
Universal laws	$p \wedge c \equiv c$	$p \vee t \equiv t$
Absorption laws	$p \wedge (p \vee q) \equiv p$	$p \vee (p \wedge q) \equiv p$
Negation of *t* and *c*	$\neg t \equiv c$	$\neg c \equiv t$

Conditional Statement

1. Let **p** and **q** be statements. The statement $p \to q$ which is read as "If **p**, then **q**" is called a conditional statement.
2. **p** is called the hypothesis and **q** is the conclusion.
3. The statement $p \to q$ has the truth value **false** when **q** has the truth value false while **p** has truth value true, otherwise it has the truth value true.

p	q	$p \to q$
1	1	1
1	0	0
0	1	1
0	0	1

Example: Write the following statement in symbolic form

If either John takes Calculus or Betty takes Sociology then Peter will takes English.

Solution

Let denote the statements as:

1. p: John takes Calculus,
2. q: Betty takes Sociology and
3. r: Peter takes English.

Then the given statement can be symbolized as follow: $p \vee q \rightarrow r$.

Example: Construct the truth table for $p \vee \neg q \rightarrow \neg p$

p	q	$\neg p$	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \rightarrow \neg p$
1	1	0	0	1	0
1	0	0	1	1	0
0	1	1	0	0	1
0	0	1	1	1	1

Exercises: Construct the truth table for the following and give your observation

1. $p \rightarrow q$ and $\neg p \vee q$
2. $\neg(p \rightarrow q)$ and $p \wedge \neg q$
3. $p \rightarrow q$ and $\neg q \rightarrow \neg p$
4. $\neg p \rightarrow \neg q$ and $q \rightarrow p$

Contrapositive, converse and inverse of a conditional statement

Contrapositive of a conditional statement

1. The contrapositive of a conditional statement $p \to q$ is a conditional statement defined as $\neg q \to \neg p$.
2. conditional statement and its contrapositive are equivalence.

Converse and inverse of a conditional statement

1. The converse of $p \to q$ is $q \to p$.
2. The inverse of $p \to q$ is $\neg p \to \neg q$.
3. Note that
   1. A conditional statement and its converse are not logically equivalent, $p \to q \not\equiv q \to p$.
   2. A conditional statement and its inverse are not logically equivalent, $p \to q \not\equiv \neg p \to \neg q$
   3. The converse and the inverse of a conditional statement are logically equivalent each other.$q \to p \equiv \neg p \to \neg q$.

Example: Give the converse , inverse and contrapositive of the statement

If this grotesque animal is a Jersey cow, then it must be spotted.

Solution

1. The converse is: If this grotesque animal is spotted, then it must be a Jersey cow.
2. The inverse is: If this grotesque animal is not a Jersey cow, then it cannot be spotted.
3. The contrapositive is: If this grotesque animal is not spotted, then it must not be a Jersey cow

Example: Write contrapositive, converse and inverse statements for: "If P is a square, then P is a rectangle"

Solution

1. The contrapositive is: "If P is not a rectangle, then P is not a square. "
2. The converse is: "If P is a rectangle, then P is a square. "
3. The inverse is: "If P is not a square, then P is not a rectangle."

Logical implication

Def

1. We say that a compound statement **p** logically implies a compound statement **q** when the compound statement $p \to q$ is a tautology.
2. The logical implication **p** implies **q** is symbolized as $p \Rightarrow q$.

Example: Construct the truth table for the following and give your observations

1. $[p \wedge (p \to q)] \to q$
2. $[(p \to q) \wedge (q \to r)] \to (p \to r)$
3. $[(p \to q) \wedge \neg q] \to \neg p$

Argument evaluation and rules of inference

Def

1. An argument is an assertion that the conjunction of several statements logically implies another statement.
2. A statement of the form $a \Rightarrow b$ which asserts that one compound statement logically implies another is sometimes called an argument of propositional logic.
3. If it is true that $a \Rightarrow b$ then we say that the argument is valid. We also say that **b** can be deduced or inferred from **a**, and that **b** is logical consequence of **a** or that **b** follows logically from **a**.
4. An argument that is not valid is said to be fallacious or to be a fallacy.
5. If the hypothesis is a conjunction of statements $a = a_1 \wedge a_2 \wedge ... \wedge a_n$ then each of the statements $a_1, a_2, ..., a_n$ is called a premise of the argument.

Rules of inference:Rule of Detachment

1. The rule of Detachment says that $p \wedge (p \rightarrow q) \Rightarrow q$.
2. This rule is known by the name of **Modus Ponens**, meaning the **method of affirming.**

Rules of inference: Hypothetical Syllogism

1. Law of Hypothetical Syllogism (Sometimes called Transitive Rule) says that $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow (p \rightarrow r)$.
2. Remark: The rule called **Modus Tollens**, meaning the method of denying, can be deduced from the rule of Modus Ponens. It says that $\neg q \wedge (p \rightarrow q) \Rightarrow \neg p$.

Example: Write in symbolic form and verify Modus Tollens

If you master the content then you will pass well the exam. You did not pass well the exam. Therefore you o not master the content.

Example

If the integer 35244 is divisible by 396, then the integer 35244 is divisible by 66.
If the integer 35244 is divisible by 66, then the integer 35244 is divisible by 3.
Therefore, if the integer 35244 is divisible by 396, then the integer 35244 is divisible by 3.
This argument illustrated the transitive rule by taking:
p: the integer 35244 is divisible by 396,
q: the integer 35244 is divisible by 66 and
r: the integer 35244 is divisible by 3.

$$(p \to q) \wedge (q \to r) \Rightarrow (p \to r)$$

Exercises

1. Show the following implications

a) $p \wedge q \Rightarrow p$ b) $p \vee q \Rightarrow p$ c) $p \Rightarrow (p \vee q)$

d) $\neg p \Rightarrow (p \rightarrow q)$ e) $q \Rightarrow (p \rightarrow q)$ f) $\neg(p \rightarrow q) \Rightarrow p$

g) $p \wedge (p \rightarrow q) \Rightarrow q$

2. Show the following equivalences

a) $p \rightarrow (q \rightarrow p) \equiv (p \rightarrow q)$ b) $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$

c) $(p \rightarrow q) \wedge (r \rightarrow q) \equiv (p \vee r) \rightarrow q$ d) $\neg(p \rightarrow q) \equiv p \wedge \neg q$

4. Give the contrapositive of the statement: "If you don't pay the school fees, you'll never see your Progress report."

Biconditional statement

1. The statement $p \leftrightarrow q$ is defined to be the statement $(p \rightarrow q) \wedge (q \rightarrow p)$.
2. For this reason, the double headed arrow $\leftrightarrow$ is called the biconditional.
3. The biconditional $p \leftrightarrow q$, which we read "**p if and only if q**" or "**p if and only if q**" is true if both **p** and **q** have the same truth values and false if **p** and **q** have opposite truth values.

p	q	$p \leftrightarrow q$
1	1	1
1	0	0
0	1	0
0	0	1

Exercises: Construct the truth table for

1. $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$
2. $p \leftrightarrow q$ and $(\neg p \vee q) \wedge (\neg q \vee p)$
3. $\neg(p \leftrightarrow q)$ and $(p \vee q) \wedge \neg(p \wedge q)$
4. $\neg(p \leftrightarrow q)$ and $(p \wedge \neg q) \vee (\neg p \wedge q)$

Predicate

Definition

A declarative sentence is an open statement or a predicate if:

1. It contains one or more variables, and
2. It is not a proposition, but
3. It becomes a proposition when the variables in it are replaced by certain allowable choices.
4. A predicate can be denoted by a small letter followed by its variables.
5. A predicate requiring m variables is called an m-place predicate.
6. An m-place predicate can be noted as $p(x_1, x_2, x_3, ..., x_m)$

Domain of interpretation

1. Domain of interpretation is the set of possible values for the variables is the predicate can take values.
2. This set is called the domain of interpretation or the universe of discourse, or often simply the domain.
3. It is commonly denoted by D.
4. This set is never empty; that is, there is always at least one element in any domain of interpretation.
5. If $p(x)$ is a predicate and x has domain D, the truth set of $p(x)$ is the set of all elements t of D such that $p(t)$ is true.

Example

Assume a predicate $p(x)$ that represents statement: "x is a prime number" from domain $D = \{2, 3, 4, 5, 6, 7\}$. Then the truth values for different x are give in the following table:

p(x)	Truth values
p(2)	1
p(3)	1
p(4)	0
p(5)	1
p(6)	0
p(7)	1

All statements $p(2), p(3), p(4), p(5), p(6), p(7)$ are propositions. But $p(x)$ with variable x is not a proposition. The truth set of this predicate is $\{2, 3, 5, 7\}$

Example

1. Let $p(x)$ be the predicate: "x is even number" from domain $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Give the truth set of $p(x)$
2. Assume a predicate $p(x, y)$ that represents the statement: $x + y = 8$. Give the truth value of $p(3, 5)$ and $p(1, 5)$.
3. Let $q(x, y, z)$ denote the statement " $x^2 + y^2 = z^2$ ".
   1. What is the truth value of $q(3, 4, 5)$?
   2. What is the truth value of $q(2, 2, 3)$?

Quantifiers

Introduction

1. Given a predicate p and a domain D, how often does the predicate become true?
2. Some of the possible answers are always, sometimes and never.
3. More generally we want to know in what quantity of case the predicate becomes true.
4. Certain words, symbols or groups of words that help to answer such a question are called quantifiers.
5. Quantifiers tell us something about "quantity".
6. **They help us to decide the frequencies with which a predicate becomes true, whether it is satisfied by no element of a domain, or one element, or some elements, or all elements**.

The most important quantifiers are

1. The existential quantifiers $\exists$ ("there exist"), and
2. The universal quantifier $\forall$ ("for all")
3. The appearance of each quantifier is signed to evoke its meaning.
4. When a variable is placed in front of a variable, we say that the variable is quantified.

Existential quantifier

1. The statement $(\exists x)$ p(x) is read as "there exists an x such that $p(x)$ is true".
2. This means that $p(x)$ is sometimes true. That is, there is at least one value a in D for which $p(a)$ is true.
3. We also write $\exists x, p(x)$ or $\exists x \in D, p(x)$. Where the second version means "there exit x in D such that $p(x)$ is true".
4. Also often used in predicate logic the unique existential quantifier $\exists!$ which means "there exists a unique a in D for which $p(a)$ is true."

Universal quantifier

1. The statement $(\forall x)$ p(x) is read as "for all x such that $p(x)$ is true".
2. This means that $p(x)$ is always true. That is no matter what value a from domain D is chosen for x, the resulting statement $p(a)$ is true.
3. We also write $\forall x, p(x)$ or $\forall x \in D, p(x)$. Where the second version means "for all x in D such that $p(x)$ is true".

EX1

1. Let $p(x)$ mean $x^2 \geq x$. Let $D_1 = \{1, 2, 3, 4....\}$ and $D_2 = \mathbb{R}$ be two domains of interpretation. The statement $(\forall x)p(x)$ is true in D_1 but false in D_2 since for example $\frac{1}{2} \in D_2$ but $(\frac{1}{2})^2 = \frac{1}{4} < \frac{1}{2}$.
2. Let $q(x)$ mean $x^2 = -x$. Let $D_1 = \mathbb{N}$ and $D_2 = \mathbb{Z}$ be two domains of interpretation. The statement $(\exists x)p(x)$ is false in D_1 but true in D_2 since for example $0 \in D_2$ but $(0)^2 = -0 = 0$.
3. Consider the predicates: $r(x) : x - 7 = 2$ and $s(x) : x > 9$. If the universe of discourse is the real numbers, give the truth value of propositions: $(\exists x)s(x) \wedge \neg(\forall x)r(x)$
   True. There exist real numbers that are greater than 9, and not all real numbers are equal to 9.

Exercises s

1. Translate the following into symbolic form:
   a) Somebody cried out for help and called the police
   b) Nobody can ignore her
2. Consider the following predicates:

   $p(x, y): x > y \qquad q(x, y): x \leq y \qquad r(x): x-7 = 2 \qquad s(x): x > 9$

   If the universe of discourse is the real numbers, give the truth value of each of the following propositions:

   a) $(\exists x) r(x)$ b) $(\forall y)[\neg s(y)]$
3. Prove that
   a) $\forall x(p(x) \wedge q(x)) \equiv \forall x p(x) \wedge \forall x q(x)$
   b) $\exists x(p(x) \vee q(x)) \equiv \exists x p(x) \vee \exists x q(x)$

Negation of quantifiers

1. Consider the following statement: **"All dogs have tails."**
2. When we negate this statement, We get the statement: **"Not all dogs have tails."**
3. But what this means is that: "There is at least one dog that does not have a tail."

Let us express these ideas symbolically

1. Let $p(x)$ mean "dog x has a tail".
2. Then all dogs have tails is represented by $\forall x, p(x)$ and
3. dog x doesn't have a tail is $\neg p(x)$.
4. So not all dogs have a tails is $\neg[\forall x, p(x)]$.
5. Abbreviating there is at least one dog that doesn't have a tail or there exists a tailed dog, we can say that there exists a tailless dog is $\exists x, \neg p(x)$

$\neg[\forall x, p(x)] \equiv \exists x, \neg p(x)$ and $\neg[\exists x, p(x)] \equiv \forall x, \neg p(x)$

Negating a universally quantified formula changes it into an existentially quantified formula and vice-versa with the part of the formula after the quantifier becoming negated.

Combination of quantifiers

Let $p(x, y)$ be a predicate involving two variables x and y. Then both variables can be quantified. Consider the following eight possibilities:If the same quantifier is used twice, then the order does not matter. So **1 and 2** have the same meaning as each other, and **3 and 4** have the same meaning as each other.

1. $\exists x, \exists y : p(x, y)$
2. $\exists y, \exists x : p(x, y)$
3. $\forall x, \forall y : p(x, y)$
4. $\forall y, \forall x : p(x, y)$
5. $\forall x, \exists y : p(x, y)$
6. $\exists y, \forall x : p(x, y)$
7. $\exists x, \forall y : p(x, y)$
8. $\forall y, \exists x : p(x, y)$

1. The expression **5** means that for each x that you specify, there exists a y which may depend in x such that $p(x, y)$ is true. So changing x may change y.
2. The expression **6** means that no matter what value we choose for x, the statement $p(x, y)$ is true.

Example

Let $p(x, y)$ be the predicate $x < y$. Let domain D be the set of all real numbers.

1. Then, the statement $\forall x, \exists y : p(x, y)$ is true. Indeed, for each value of x we try to find a value of y such that $x < y$. We can choose y to be $x + 1$.
2. However, the statement $\exists y, \forall x : p(x, y)$ is false because it means that there is some constant value of y which is greater than real number x.

Exercises

Consider the following predicates:

$p(x, y): x > y \qquad q(x, y): x \leq y$

$r(x): x - 7 = 2 \qquad s(x): x > 9$

If the universe of discourse is the real numbers, give the truth value of each of the following propositions:

a) $(\forall x)(\exists y)\, p(x, y)$

b) $(\exists y)(\forall x)\, q(x, y)$

c) $(\forall x)(\forall y)\left[p(x, y) \vee q(x, y)\right]$

d) $(\exists y)(\forall x)\left[s(y) \wedge q(x, y)\right]$

Argument with quantified statements

Universal instantiation

The rule of universal instantiation says that if some property is true of everything in a domain, then it is true of any particular thing in the domain. $\forall x$ if $p(x)$. Therefore $p(a)$.

Example

All human beings are mortal.
Socrates is a human being.
Therefore, Socrates is mortal

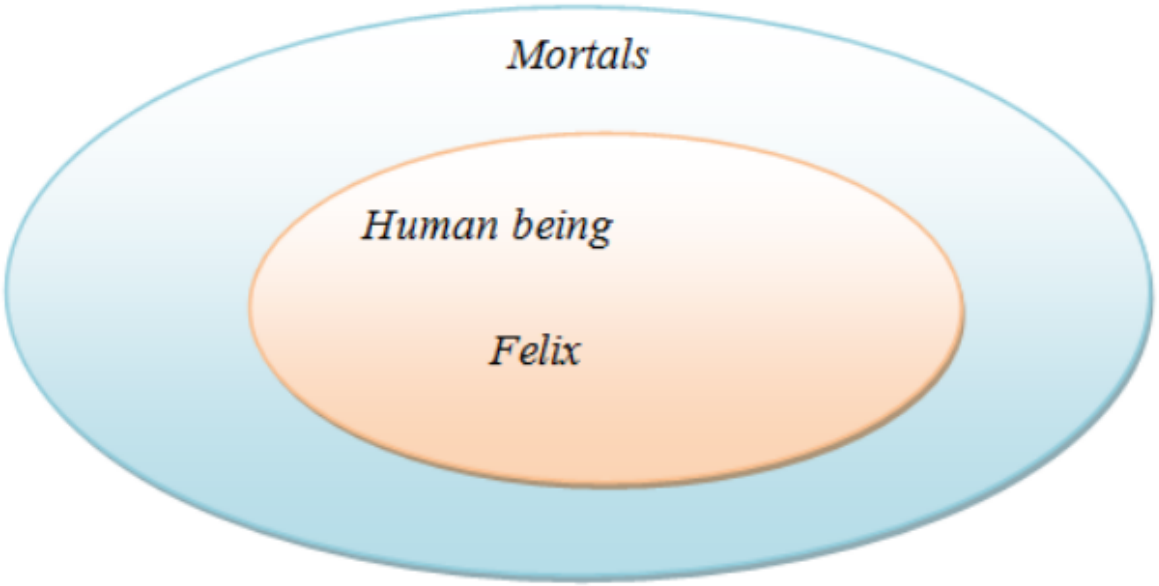
Universal Modus Ponens

1. The rule of universal instantiation can be combined with the modus ponens to obtain the rule called "Universal Modus Ponens." $\forall x$ if $p(x)$ then $q(x)$. p (a) for a particular a, therefore q (a)
2. Means, If x makes $p(x)$ true, then x makes $q(x)$ true. a makes $p(x)$ true, therefore a makes $q(x)$ true.
3. The argument from above consists of two premises and a conclusion and at least one premise is quantified. An argument of this form is called a **syllogism**.

All human beings are mortal.

Felix is a human being.

Therefore, Felix is mortal.



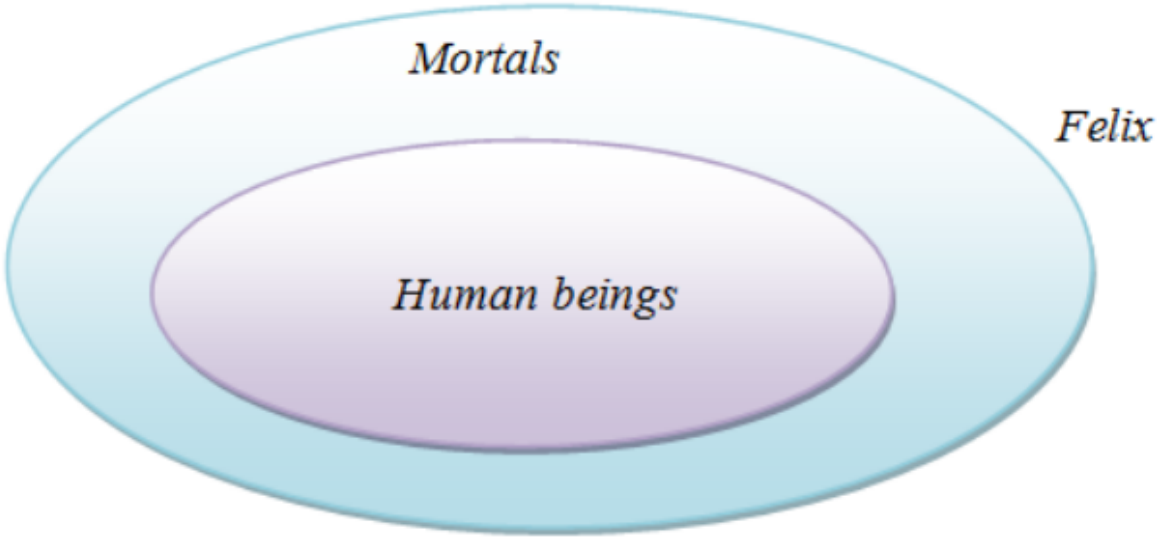
Universal Modus Tollens

1. The rule of universal instantiation can be combined with the modus Tollens to obtain the rule called "Universal Modus Tollens." $\forall x$ if $p(x)$ then $q(x)$. $\neg q(a)$ for a particular a, therefore $\neg p(a)$
2. Means, If x makes $p(x)$ true, then x makes $q(x)$ true. a makes $q(x)$ not true, therefore a makes $p(x)$ not true.

All human beings are mortal.

Felix is not mortal.

Therefore, Felix is not human being.



Converse and inverse errors

Converse error

The following argument is **invalid**:

$\forall x$, if $p(x)$ then $q(x)$

$q(a)$ for a particular a, therefore $p(a)$

Inverse error

The following argument is **invalid:**

$\forall x$, if $p(x)$ then $q(x)$

$\neg p(a)$ for a particular a, therefore $\neg q(a)$

similarities of operations in set theory and logic

The following table summarizes similarities of operations in set theory and logic

Let p, q and r be any propositions and A, B and C 3 sets

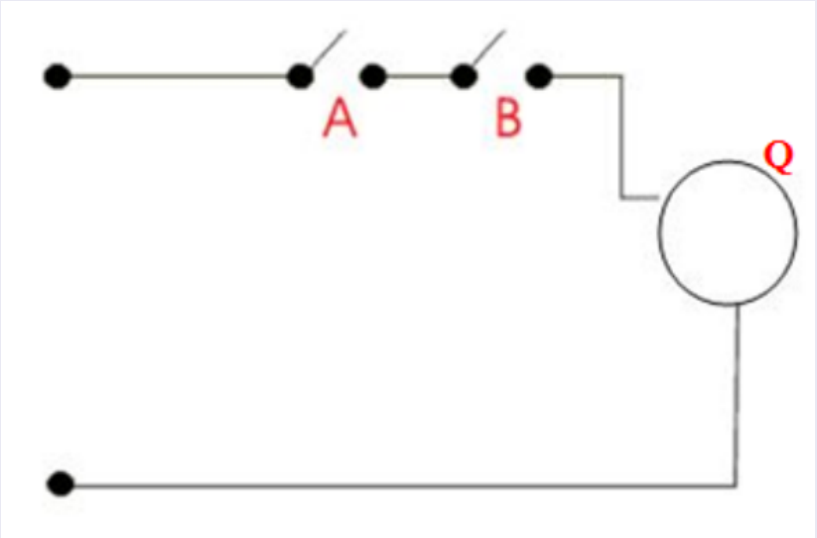
Logic	Set theory	Law
$p \wedge q \equiv q \wedge p$	$A \cap B = B \cap A$	Commutative
$p \wedge p \equiv p$	$A \cap A = A$	Idempotent
$p \vee q \equiv q \vee p$	$A \cup B = B \cup A$	Commutative
$p \vee p \equiv p$	$A \cup A = A$	Idempotent
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(A \cap B) \cap C = A \cap (B \cap C)$	Associative
$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(A \cup B) \cup C = A \cup (B \cup C)$	Associative
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$	Distributive
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$A \cup (B \cap C) \equiv (A \cup B) \cap (A \cup C)$	Distributive
$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$(A \cap B)' = A' \cup B'$	De Morgan's
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$(A \cup B)' = A' \cap B'$	De Morgan's

Applications

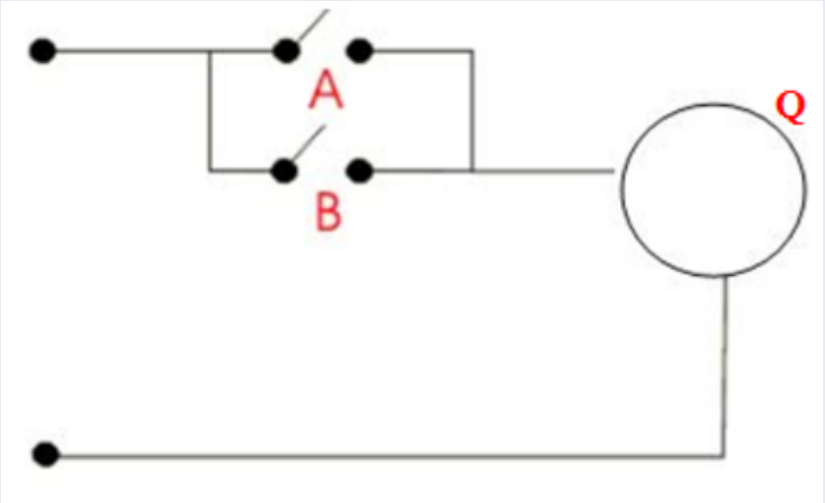
Electrical circuit and logic gates

1. The simplified **AND gate** has two inputs, **switch A** and **switch A** . The bulb will only light if **both switches are closed**. This will allow current to flow through the bulb, illuminating the filament.
2. The simplified **OR gate** two inputs, **switch A** and **switch B**. The bulb will light if either switch **A or B are closed**. This will allow current to flow through the bulb, illuminating the filament.

The simplified **AND gate**



The simplified **OR gate**



Exercises

1. *Which of the following are well formed propositional formulas?*

   1. $\vee pq$
   2. $(\neg(p \to (q \wedge p)))$
   3. $(\neg(p \to (q = p)))$
   4. $(\neg(\Diamond(q \vee p)))$
   5. $(p \wedge \neg q) \vee (q \to r)$
   6. $p \neg r$

2

Let's consider the interpretation v where $v(p) = F$, $v(q) = T$, $v(r) = T$. Does v satisfy the following propositional formulas?

1. $(p \to \neg q) \vee \neg(r \wedge q)$
2. $(\neg p \vee \neg q) \to (p \vee \neg r)$
3. $\neg(\neg p \to \neg q) \wedge r$
4. $\neg(\neg p \to q \wedge \neg r)$

3

Compute the truth table of $(F \vee G) \wedge \neg(F \wedge G)$.

Solution.

F	G	$F \vee G$	$F \wedge G$	$\neg(F \wedge G)$	$(F \vee G) \wedge \neg(F \wedge G)$
T	T	T	T	F	**F**
T	F	T	F	T	**T**
F	T	T	F	T	**T**
F	F	F	F	T	**F**

4

Use the truth tables method to determine whether $(p \to q) \vee (p \to \neg q)$ *is valid.*

Solution.

p	q	$p \to q$	$\neg q$	$p \to \neg q$	$(p \to q) \vee (p \to \neg q)$
T	T	T	F	F	**T**
T	F	F	T	T	**T**
F	T	T	F	T	**T**
F	F	T	T	T	**T**

The formula is valid since it is satisfied by every interpretation.

5 Find the truth tables of each of the following:

$(\neg p \vee q) \wedge (q \to \neg r \wedge \neg p) \wedge (p \vee r)$, $p \wedge \neg q \to p \wedge q$

- $(p \to p) \to p$
- $p \to (p \to p)$
- $p \vee q \to p \wedge q$
- $p \vee (q \wedge r) \to (p \wedge r) \vee q$
- $p \to (q \to p)$
- $(p \wedge \neg q) \vee \neg(p \leftrightarrow q)$

6

Use the truth table method to verify whether the following formulas are valid, satisfiable or unsatisfiable:

- $(p \to q) \wedge \neg q \to \neg p$
- $(p \to q) \to (p \to \neg q)$
- $(p \vee q \to r) \vee p \vee q$
- $(p \vee q) \wedge (p \to r \wedge q) \wedge (q \to \neg r \wedge p)$
- $(p \to (q \to r)) \to ((p \to q) \to (p \to r))$
- $(p \vee q) \wedge (\neg q \wedge \neg p)$
- $(\neg p \to q) \vee ((p \wedge \neg r) \leftrightarrow q)$
- $(p \to q) \wedge (p \to \neg q)$
- $(p \to (q \vee r)) \vee (r \to \neg p)$

7

Use the truth tables method to determine whether the formula $\varphi : p \wedge \neg q \to p \wedge q$ *is a logical consequence of the formula* $\psi : \neg p$.

Solution.

p	q	$\neg p$	$p \wedge \neg q$	$p \wedge q$	$p \wedge \neg q \to p \wedge q$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	**T**	F	F	**T**
F	F	**T**	F	F	**T**

$\psi \models \varphi$ *since each*

interpretation satisfying psi *satisfies also* φ.

8

Use the truth table method to verify whether the following logical consequences and equivalences are correct:

- $(p \to q) \models \neg p \to \neg q$
- $(p \to q) \wedge \neg q \models \neg p$
- $p \to q \wedge r \models (p \to q) \to r$
- $p \vee (\neg q \wedge r) \models q \vee \neg r \to p$
- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- $(p \vee q) \wedge (\neg p \to \neg q) \equiv q$

- $(p \vee q) \wedge (\neg p \rightarrow \neg q) \equiv q$
- $(p \wedge q) \vee r \equiv (p \rightarrow \neg q) \rightarrow r$
- $(p \vee q) \wedge (\neg p \rightarrow \neg q) \equiv p$
- $((p \rightarrow q) \rightarrow q) \rightarrow q \equiv p \rightarrow q$

9

Let's consider a propositional language where

- p *means* "Paola is happy",
- q *means* "Paola paints a picture",
- r *means* "Renzo is happy".

Formalize the following sentences:

1. "if Paola is happy and paints a picture then Renzo isn't happy"
2. "if Paola is happy, then she paints a picture"
3. "Paola is happy only if she paints a picture"

Solution.

1. $p \wedge q \rightarrow \neg r$
2. $p \rightarrow q$
3. $\neg(p \wedge \neg q)$ *..which is equivalent to* $p \rightarrow q$

10

Let's consider a propositional language where

- p *means* "x is a prime number",
- q *means* "x is odd".

Formalize the following sentences:

1. "x being prime is a sufficient condition for x being odd"
2. "x being odd is a necessary condition for x being prime"

Solution. *1. and 2.* $p \to q$

11

Let $A =$"Aldo is Italian" *and* $B =$"Bob is English".
Formalize the following sentences:

1. "Aldo isn't Italian"
2. "Aldo is Italian while Bob is English"
3. "If Aldo is Italian then Bob is not English"
4. "Aldo is Italian or if Aldo isn't Italian then Bob is English"
5. "Either Aldo is Italian and Bob is English, or neither Aldo is Italian nor Bob is English"

Solution.

1. $\neg A$
2. $A \wedge B$
3. $A \rightarrow \neg B$
4. $A \vee (\neg A \rightarrow B)$ *logically equivalent to* $A \vee B$
5. $(A \wedge B) \vee (\neg A \wedge \neg B)$ *logically equivalent to* $A \leftrightarrow B$

12

Let's consider a propositional langiage where

- $A =$"Angelo comes to the party",
- $B =$"Bruno comes to the party",
- $C =$"Carlo comes to the party",
- $D =$"Davide comes to the party".

Formalize the following sentences:

Formalize the following sentences:

1. "If Davide comes to the party then Bruno and Carlo come too"
2. "Carlo comes to the party only if Angelo and Bruno do not come"
3. "Davide comes to the party if and only if Carlo comes and Angelo doesn't come"
4. "If Davide comes to the party, then, if Carlo doesn't come then Angelo comes"
5. "Carlo comes to the party provided that Davide doesn't come, but, if Davide comes, then Bruno doesn't come"
6. "A necessary condition for Angelo coming to the party, is that, if Bruno and Carlo aren't coming, Davide comes"
7. "Angelo, Bruno and Carlo come to the party if and only if Davide doesn't come, but, if neither Angelo nor Bruno come, then Davide comes only if Carlo comes"

Solution.

1. $D \to B \wedge C$
2. $C \to \neg A \wedge \neg B$
3. $D \leftrightarrow (C \wedge \neg A)$
4. $D \to (\neg C \to A)$
5. $(\neg D \to C) \wedge (D \to \neg B)$
6. $A \to (\neg B \wedge \neg C \to D)$
7. $(A \wedge B \wedge C \leftrightarrow \neg D) \wedge (\neg A \wedge \neg B \to (D \leftrightarrow C))$

13 Socrate says: "If I'm guilty, I must be punished; I'm guilty. Thus I must be punished." Is the argument logically correct?

Solution

The argument is logically correct: if p means "I'm guilty" and q means "I must be punished", then:$(p \to q) \wedge p \Rightarrow q$ (modus ponens)

14 Socrate says: "If I'm guilty, I must be punished; I'm not guilty. Thus I must not be punished. Is the argument logically correct?

Solution

The argument is not logically correct: if p means "I'm guilty" and q means "I must be punished", then:$(p \to q) \wedge \neg p \not\Rightarrow \neg p$

15 Suppose we know that:

- "if Paolo is thin, then Carlo is not blonde or Roberta is not tall"
- "if Roberta is tall then Sandra is lovely"
- "if Sandra is lovely and Carlo is blonde then Paolo is thin"
- "Carlo is blonde"

Can we deduce that "Roberta is not tall" ?

16 Let $L(x, y)$ be the statement x loves y where the domain for both x and y consists of all people in your class. Use quantifiers to express each o the following:

1. Everybody loves Mary.
2. Nobody loves James.
3. Everybody loves somebody.
4. There is somebody whom everybody loves.
5. Nobody loves everybody.
6. Everyone loves himself or herself.

17 Let $W(x, y)$ mean that student x has visited website y, where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.

1. $\exists y \forall z (y \neq David) \wedge [W(David, z) \rightarrow W(y, z)]$
2. $\exists y [W(John, y) \wedge W(Mary, y)]$
3. $\exists y [W(John, y) \vee W(Mary, y)]$

18 Let $M(x, y)$ be "x has sent y an e-mail message" and $T(x, y)$ be "x has telephoned y," where the domain consists for all students in your class. Use quantifiers to express each of these statements.

1. There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.
2. Every student in the class has either received an e-mail message or received a telephone call from another student in the class.

19 Negate each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

1. $\forall x \forall y P(x, y) \vee \exists x \exists y \neg Q(x, y)$
2. $\neg[\forall x \exists y P(x) \wedge \forall x \forall y \neg Q(x, y)]$

20 Let $P(x)$ be the predicate $P(x) \equiv \frac{x^4-1}{x} \geq 0$:

1. Write $P(2), P(1/2), P(-1), P(-1/2)$, and $P(-8)$, and indicate which of these statements are true and which are false.
2. Find the truth set of P(x) if the domain of x is the set of all real numbers.
3. If the domain is the set of all positive real numbers, what is the truth set of P(x)?.

Q1.

Let $L(x, y)$ be the statement "x loves y," where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.

a) Everybody loves Jerry.

$\forall x L(x, Jerry)$

b) Everybody loves somebody.

$\forall x \exists y L(x, y)$

c) There is somebody whom everybody loves.

$\exists y \forall x L(x, y)$

d) Nobody loves everybody.

$\forall x \exists y \neg L(x, y)$ or $\neg \exists x \forall y L(x, y)$

i Everyone loves himself or herself

$\forall x L(x, x)$

Q2.

Let $W(x, y)$ mean that student x has visited website y, where the domain for x consists of all students in your school and the domain for y consists of all websites. Express each of these statements by a simple English sentence.

d $\exists y(W(\text{Ashok Puri}, y) \wedge W(\text{Cindy Yoon}, y))$

There is a website that both Ashok and Cindy have visited.

e $\exists y \forall z(y \neq (\text{David Belcher}) \wedge (W(\text{David Belcher}, z) \rightarrow W(y, z)))$

There is a person besides David who has visited all the websites that David has visited.

f $\exists x \exists y \forall z(((x \neq y) \wedge (W(x, z) \leftrightarrow W(y, z))))$

There are two distinct people who have visited exactly the same sites.

Q3.

Let $M(x, y)$ be "x has sent y an e-mail message" and $T(x, y)$ be "x has telephoned y," where the domain consists for all students in your class. Use quantifiers to express each of these statements.

k There is a student in your class who has not received an e-mail message from anyone else in the class and who has not been called by any other student in the class.

$\exists x \forall y((x \neq y) \rightarrow (\neg M(y, x) \wedge \neg T(y, x)))$

l Every student in the class has either received an e-mail message or received a telephone call from another student in the class.

$\forall x \exists y((x \neq y) \wedge (M(y, x) \vee T(y, x)))$

m There are at least two students in your class such that one student has sent the other e-mail and the second student has telephoned the first student

$\exists x \exists y((x \neq y) \wedge (M(x, y) \wedge T(y, x)))$

Q4.

Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

a) $\neg\forall x\forall y P(x,y)$

$$\begin{aligned}
&\neg\forall x\forall y P(x,y) \\
&\equiv \exists x\neg\forall y P(x,y) && \text{De Morgan's laws for quantifiers} \\
&\equiv \exists x\exists y\neg P(x,y) && \text{De Morgan's laws for quantifiers}
\end{aligned}$$

d $\neg(\exists x\exists y\neg P(x,y) \wedge \forall x\forall y(Q(x,y))$

$$\begin{aligned}
&\neg(\exists x\exists y\neg P(x,y) \wedge \forall x\forall y(Q(x,y)) \\
&\equiv (\neg\exists x\exists y\neg P(x,y)) \vee (\neg\forall x\forall y Q(x,y)) && \text{De Morgan's Law} \\
&\equiv (\forall x\neg\exists y\neg P(x,y)) \vee (\neg\forall x\forall y Q(x,y)) && \text{De Morgan's laws for quantifiers} \\
&\equiv (\forall x\forall y\neg\neg P(x,y)) \vee (\neg\forall x\forall y Q(x,y)) && \text{De Morgan's laws for quantifiers} \\
&\equiv (\forall x\forall y P(x,y)) \vee (\neg\forall x\forall y Q(x,y)) && \text{Double Negation} \\
&\equiv (\forall x\forall y P(x,y)) \vee (\exists x\neg\forall y Q(x,y)) && \text{De Morgan's laws for quantifiers} \\
&\equiv (\forall x\forall y P(x,y)) \vee (\exists x\exists y\neg Q(x,y)) && \text{De Morgan's laws for quantifiers}
\end{aligned}$$

Boolean algebra

Definition:

1. The algebra of logic is a Boolean algebra. It works with variables with two different values, such as 0 (False) and 1 (True), as well as logically significant operations.
2. George Boole invented the first way of manipulating symbolic logic, which later became known as Boolean Algebra.
3. Because of its vast applications in switching theory, developing basic electronic circuits, and designing digital computers, Boolean algebra has become a vital tool in computer science.

The following are the basic operations of Boolean algebra:

- AND or conjunction operation
- OR operation or disjunction
- Not or Negative operation

Rules:

Suppose X and Y are two Boolean variables, then the three operations are defined as follows;

1. X AND Y, satisfies $X \wedge Y = True$, if $X = Y = True$ or else $X \wedge Y = False$.
2. X OR Y, satisfies $X \vee Y = False$, if $X = Y = False$, else $X \vee Y = True$.
3. $\neg X$ satisfies $\neg X = False$, if $X = True$ and $\neg X = True$ if $X = False$

Boolean Identities

Double Complement Law	$\sim(\sim X) = X$
Complement Law	$X+\sim X = 1$ (OR Form) $X.\sim X = 0$ (AND Form)
Idempotent Law	$X+X = X$ (OR Form) $X.X = X$ (AND Form)
Identity Law	$X+0 = X$ (OR Form) $X.1 = X$ (AND Form)
Dominance Law	$X+1 = 1$ (OR Form) $X.0 = 0$ (AND Form)
Commutative Law	$X+Y = Y+X$ (OR Form) $X.Y = Y.X$ (AND Form)
Associative Law	$X+(Y+Z) = (X+Y)+Z$ (OR Form) $X.(Y.Z) = (X.Y).Z$ (AND Form)

Associative Law	X+(Y+Z) = (X+Y)+Z (OR Form) X.(Y.Z) = (X.Y).Z (AND Form)
Absorption Law	X.(X+Y) = X X+(X.Y) = X
Simplification Law	X.(~X+Y) = X.Y X+(~X.Y) = X+Y
Distributive Law	X+(Y.Z) = (X+Y).(X+Z) X.(Y+Z) = (X.Y)+(X.Z)
De-Morgan's Law	~(X.Y)=~X+~Y ~(X+Y)=~X.~Y

Example with solution

Using Boolean identities, reduce the given Boolean expression:

$F(X, Y, Z) = X'Y + YZ' + YZ + XY'Z'$

Solution:

Given, $F(X, Y, Z) = X'Y + YZ' + YZ + XY'Z'$

Using the idempotent law, we can write $YZ' = YZ' + YZ'$

$\Rightarrow F(X, Y, Z) = X'Y + (YZ' + YZ') + YZ + XY'Z'$

Now, interchange the second and third term, we get

$\Rightarrow F(X, Y, Z) = X'Y + (YZ' + YZ) + (YZ' + XY'Z')$

By using distributive law,

$\Rightarrow F(X, Y, Z) = X'Y + Y(Z' + Z) + Z'(Y + XY')$

Using $Z' + Z = 1$ and absorption law $(Y + XY') = (Y + X)$,

$\Rightarrow F(X, Y, Z) = X'Y + Y.1 + Z'(Y + X)$

$\Rightarrow F(X, Y, Z) = X'Y + Y + Z'(Y + X)$ [Since $Y.1 = Y$]

$\Rightarrow F(X, Y, Z) = Y(X' + 1) + Z'(Y + X)$

$\Rightarrow F(X, Y, Z) = Y.1 + Z'(Y + X)$ [As $(X' + 1) = 1$]

$\Rightarrow F(X, Y, Z) = Y + Z'(Y + X)$ [As, $Y.1 = Y$]

$\Rightarrow F(X, Y, Z) = Y+YZ'+XZ'$

$\Rightarrow F(X, Y, Z) = Y(1+Z')+XZ'$

$\Rightarrow F(X, Y, Z) = Y.1+XZ'$ [Since $(1 + Z') = 1$]

$\Rightarrow F(X, Y, Z) = Y+XZ'$ [Since $Y.1 = Y$]

Hence, the simplified form of the given Boolean expression is $F(X, Y, Z) = Y+XZ'$.

Reduce the following Boolean expression: $F(P, Q, R) = (P + Q)(P + R)$

Solution:

Given, F(P ,Q, R)=(P+Q)(P+R)

Using distributive law,

⇒ F(P, Q, R) = P.P + P.R +Q.P + Q.R

Using Idempotent law,

⇒ F(P, Q, R) = P + P.R +Q.P + Q.R

Again using distributive law, we get

⇒ F(P, Q, R) = P(1+R) + Q.P + Q.R

Using dominance law, we can write

⇒ F(P, Q, R) = P + Q.P + Q.R

Again using distributive law, we get

⇒ F(P, Q, R) = (P+1).P+ Q.R

Therefore, using dominance law, we can get the reduced form as follows:

$\Rightarrow F(P, Q, R) = 1.P+Q.R$

$\Rightarrow F(P, Q, R) = P+Q.R$

Hence, the reduced form of $F(P, Q, R) = (P+Q)(P+R)$ is $F(P, Q, R) = P+Q.R$.

What is the equivalent expression for the Boolean expression $x'y'z + yz + xz$?

Given Boolean expression: $x'y'z + yz + xz$

$x'y'z + yz + xz = z(x'y'+y+x)$

Now, apply distributive law for the first two terms inside the bracket.

$x'y'z + yz + xz = z[(x'+y)\ (y+y')+ x]$

$x'y'z + yz + xz = z\ [(x' + y)\ .\ 1 + x]$ [Since $A+A' = 1$]

$x'y'z + yz + xz = z\ [x' + y + x]$

Further $x+x' = 1$

So, $x'y'z + yz + xz = z(1+y)$

Now, using null law, $1+y = 1$

$x'y'z + yz + xz = z.1$

Now, using identity law, $A.1 = A$

Therefore, $x'y'z + yz + xz = z$.

Hence, the Boolean expression equivalent to $x'y'z + yz + xz$ is z.

What is the simplified sum of product form for the Boolean expression: $(A + B' + C')(A + B' + C)(A + B + C')$

Solution:

Given Boolean expression: $(A + B' + C')(A + B' + C)(A + B + C')$

Now, using the distributive law, we can write

$(A + B' + C')(A + B' + C)(A + B + C') = [(A+B') + C'C](A + B + C')$

Using $AA' = 0$, it becomes

$(A + B' + C')(A + B' + C)(A + B + C') = [(A+B') + 0](A + B + C')$

Since A + 0 = A,

(A + B′ + C′)(A + B′ + C)(A + B + C′) = (A+B′)(A + B + C′)

(A + B′ + C′)(A + B′ + C)(A + B + C′) = A + B′.(B+C′)

Again, using distributive law

(A + B′ + C′)(A + B′ + C)(A + B + C′) = A + B′B +B′C′

(A + B′ + C′)(A + B′ + C)(A + B + C′) = A + 0 + B′C′

(A + B′ + C′)(A + B′ + C)(A + B + C′) = A + B′C′.

Reduce the Boolean expression: $A = XY + X(Y + Z) + Y(Y + Z)$

Solution:

Given: A = XY + X(Y+Z) + Y(Y+Z)

The given Boolean expression can be written as follows:

A = XY + XY + XZ + YY + YZ

Using Idempotent law, Y.Y = Y

Thus, A = XY + XY + XZ + Y + YZ

Again, XY + XY = XY, we get

A = XY + XZ + Y + YZ

A = XY + XZ + Y(1+Z)

Using null law, (i.e., $1+A = 1$), we can write

$A = XY + XZ + Y.1$

Using Identity law, $1.Y = Y$.

So, $A = XY + XZ + Y$

Now, the above form can be written as:

$A = Y(1+X) + XZ$

$A = Y.1 + XZ$

$A = Y + XZ$

Hence, the reduced form of the Boolean expression $A = XY + X(Y+Z) + Y(Y+Z)$ is $A = Y+XZ$.

Check whether $AC + ABC = AC$ is true or false?

Solution:

Given Boolean expression: AC + ABC = AC

AC (1+B) = AC

Now, using the null law 1+B = 1, the above expression can be written as:

AC.1 = AC

Thus, using the identity law, 1.A = A, the above expression becomes:

AC = AC

Hence, AC + ABC = AC is true.

Consider a Boolean set $B = \{0, 1\}$ and two variables x and y in B to compete the following table

x	y	$x+y$	$(x+y)'$	xy
0	0			
0	1			
0	0			
1	1			
1	0			
1	1			

x	y	$x+y$	$(x+y)'$	xy
0	0	0	1	0
0	1	1	0	0
0	0	0	1	0
1	1	1	0	1
1	0	1	0	0
1	1	1	0	1

THANK YOU VERY MUCH!!